

Re-implementation of InteractionMesh-Based Close Interaction Motion Adaptation with a Preliminary DNN-Based Solver

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Abstract

Close interaction motion adaptation aims to retarget motions involving multiple interacting characters while preserving spatial relationships between them. Unlike conventional motion retargeting, this problem requires maintaining both individual motion plausibility and interaction structure, making it particularly challenging under changes in character proportions.

This project investigates a hybrid approach that combines constraint-based optimization with a learning-based solver. A complete pipeline is constructed, starting from BVH motion data, followed by skeleton reconstruction, InteractionMesh construction, and formulation of a constrained adaptation problem. A dataset is generated by varying bone-length scaling factors, and a neural network is trained to predict adapted poses within a constrained optimization framework inspired by DC3.

The results show that the proposed system is able to preserve the overall interaction structure under moderate scaling changes. However, quantitative analysis reveals that geometric constraints, particularly bone-length consistency, are not fully satisfied, especially under stronger asymmetric deformations. These findings suggest that while learning-based methods can approximate constrained motion adaptation, additional mechanisms are required to ensure reliable constraint satisfaction. Overall, the project demonstrates the feasibility of integrating neural networks into constrained motion adaptation and identifies key challenges for future work.

Contents

1	Introduction	5
1.1	Introduction	5
1.2	Chapter Organization	7
2	Background	8
2.1	Character Motion Retargeting	8
2.2	Close Interaction Motion Adaptation	8
2.3	InteractionMesh and Spatial Relationship Preservation	10
2.4	Motion Representation and BVH-Based Skeleton Data	11
2.5	Learning-Based Solvers for Constrained Optimization	12
2.6	DC3 and Its Relevance to This Project	13
2.7	Dataset Source and Position of This Project	14
3	System Design and Methodology	15
3.1	Overall Pipeline Overview	15
3.2	BVH Parsing and Motion Representation	16
3.3	InteractionMesh Construction	17
3.4	Constrained Motion Adaptation Formulation	19
3.4.1	Deformation Energy	20
3.4.2	Soft Constraints	21
3.4.3	Hard Constraints	22
3.4.4	Equality Completion for the Neural Solver	22
3.5	Dataset Construction	23
3.5.1	Reference Frame and Base Formulation	24
3.5.2	Construction of Equality Constraints	24
3.5.3	Instance Generation by Bone-Length Scaling	25
3.5.4	Inequality Constraints	25
3.5.5	Input and Output of the Neural Solver	26
3.6	Preliminary DNN-Based Solver	26

3.6.1	Network Architecture	27
3.6.2	Training Objective	27
3.6.3	Role of Correction	27
3.7	Prediction and Visualization Workflow	28
3.7.1	Construction of a New Test Instance	28
3.7.2	Forward Prediction	29
3.7.3	Numerical Checking	29
3.7.4	Export to Blender and Qualitative Inspection	30
3.7.5	Role in the Overall Pipeline	30
3.8	Implementation Scope and Current Limitations	30
4	Results and Evaluation	32
4.1	Evaluation Setup	32
4.2	Qualitative Results	33
4.3	Bone-Length Scaling Analysis	34
4.4	Discussion of Failure Cases and Limitations	35
5	Conclusion and Evaluation	36

1 Introduction

1.1 Introduction

Close interactions between body parts of one or multiple characters, such as dancing, wrestling, sword fighting, or moving within a restricted environment, are common in character animation and computer games, making them an important class of motions for animation systems to handle [6]. Unlike conventional motion retargeting (shown in Figure 1), close interaction motion adaptation must preserve not only the plausibility of each individual character’s motion, but also the spatial relationships between interacting body parts, since these relationships are closely tied to the semantics of the scene [6, 8].

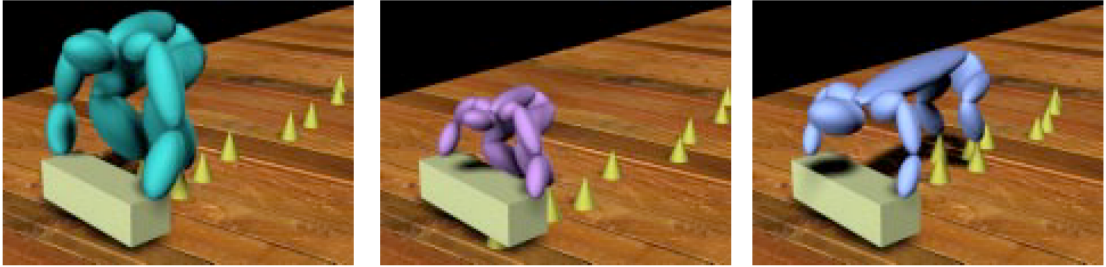


Figure 1: Differently sized characters pick up an object. Same scene semantics is preserved while the positions of characters are determined by the object. **Source:** Adapted from [3].

This makes the problem substantially more difficult than standard motion retargeting. Traditional motion representations are typically based on joint angles and explicit kinematic constraints, yet many interaction cues are implicit and therefore difficult to encode directly. As a result, adapting motions to characters with different sizes or morphologies can easily alter the original interaction context [6, 3]. For this reason, close interaction motion adaptation remains an important problem in computer animation. Classical retargeting methods provide a strong constraint-based foundation for adapting motion to new characters, while later work on spatial relationship preservation shows that successful adaptation of interaction motions depends on maintaining the geometric context between characters rather than simply transferring poses frame by frame [6, 8, 3].

The technical challenge follows naturally from this requirement. Close interaction motion adaptation is difficult because the goal is not only to retarget motion to a new character, but also to preserve the spatial relationship between two interacting characters over time [6]. Many important interaction cues, such as close proximity, collision avoidance, and coordinated body movement, are implicit. Standard motion

representations, which usually rely on joint angles and a limited number of explicit constraints, are therefore insufficient to fully describe these behaviours [6]. For this reason, optimization-based methods are often required to enforce positional, geometric, and collision-related constraints during adaptation, but these methods are typically iterative and computationally expensive [6]. At the same time, although neural networks offer a potentially faster alternative, directly predicting adapted motion with a deep model does not naturally guarantee feasibility, especially when hard constraints must be satisfied [2]. Overall, the problem is difficult because it requires both efficient motion generation and reliable preservation of interaction structure under strict constraints [6, 2].

Against this background, the present project is motivated by the observation that existing optimization-based methods provide a principled way to preserve spatial relationships in close interaction motion adaptation, but remain relatively complex to implement and expensive to solve iteratively [6]. The project therefore begins with a re-implementation of the core InteractionMesh-based pipeline, with the aim of building a complete engineering workflow from raw BVH motion data to a constrained motion adaptation formulation [6]. A second motivation is to explore whether this optimization problem can be partially accelerated by a neural network-based solver. To this end, the project draws on the idea of DC3, which combines neural prediction with constraint completion and correction for optimization problems with hard constraints [2]. In practice, the project focuses on constructing the motion representation, formulating the relevant constraints, and integrating these components into a preliminary DNN-based pipeline for predicting adapted poses. However, the project does not claim to fully solve close interaction motion adaptation. Instead, it aims to evaluate the feasibility of combining an InteractionMesh-based formulation with constrained neural prediction, while identifying the remaining limitations that prevent full constraint satisfaction and robust motion adaptation.

The completed work can be divided into four stages.

- The first stage concerns data and motion representation. BVH files were parsed to recover the skeleton hierarchy and frame-wise joint positions, which were then converted into a representation suitable for constructing an InteractionMesh-style model of two-character interaction.
- The second stage concerns formulation of the constrained adaptation problem. This included rebuilding the main geometric and optimization components needed for adaptation, such as interaction-aware structures, positional constraints, bone-length related constraints, and collision-related terms, following the InteractionMesh-based formulation of close interaction adaptation [6].

- The third stage is a preliminary integration with a neural network solver. Inspired by DC3, the project reformulates the adaptation task into a constrained learning pipeline, in which the network predicts candidate outputs that are subsequently handled using constraint-aware procedures [2].
- The fourth stage concerns result generation and qualitative inspection. The predicted motions were visualized to assess whether the adapted outputs still retained recognizable interaction structure under changed character configurations.

These completed stages lead to a working prototype rather than a fully successful final system. While it was possible to generate outputs that preserved some interaction characteristics, the results did not perfectly enforce all target constraints, particularly those related to exact body proportions and bone lengths. The completed work is therefore best characterised as a re-implementation of the original pipeline together with an initial proof-of-concept for constrained DNN-based acceleration.

The value of the project should be understood in this context. Its first contribution lies in engineering reproduction: the project reconstructs the full pipeline needed to move from BVH motion data to an InteractionMesh-style constrained adaptation problem, thereby turning the original idea in the literature into a concrete and inspectable implementation [6]. Its second contribution lies in feasibility testing: by integrating the adaptation formulation with a constrained neural solver inspired by DC3, the project provides an initial test of whether optimization-based close interaction adaptation can be partially handled by a learning-based method [2]. Although the final outputs do not perfectly satisfy all intended constraints, they still preserve some recognizable interaction structure. This suggests that the proposed integration direction is meaningful, even though substantial technical difficulties remain. The project is therefore valuable not because it completely solves the problem, but because it both establishes the necessary pipeline and reveals the main obstacles that must be addressed in future attempts to make the solver reliable.

1.2 Chapter Organization

The rest of this report is structured as follows. Chapter 2 introduces the background and related work, including close interaction motion adaptation, InteractionMesh, and constrained learning methods relevant to this project. Chapter 3 describes the design and implementation of the system, covering motion representation, constrained formulation, dataset preparation, and the preliminary DNN-based solver.

Chapter 4 presents the experimental results and evaluates the strengths and limitations of the current pipeline. Finally, Chapter 5 concludes the report and discusses possible future work.

2 Background

2.1 Character Motion Retargeting

Character motion retargeting is a fundamental problem in computer animation, since captured or designed motions often need to be transferred to new characters with different body proportions, morphologies, or kinematic structures while preserving the essential features of the original movement [3].

Early work on motion retargeting formulated this task as a constraint-based adaptation problem, where the objective is not to copy joint trajectories directly, but to preserve those motion properties that are important to the perceived meaning and functionality of the original animation [3]. This makes retargeting especially useful when the source and target characters differ in size or shape, because naive transfer of joint angles can easily distort posture, contact, or task-related behaviour [3, 6].

More generally, motion retargeting addresses the practical need to reuse existing motion data efficiently. Instead of creating a new animation sequence from scratch for every new character, retargeting methods aim to adapt a reference motion to a new body while maintaining realism and reducing manual animation effort [3]. This is important in a wide range of applications, including character animation, interactive systems, and robot motion adaptation in constrained settings [3, 5].

However, most conventional retargeting methods are primarily designed for single-character motion or for scenarios in which the key constraints can be specified explicitly. As later work has shown, this becomes insufficient when the motion involves rich spatial relationships between multiple bodies or between a body and its surrounding environment [6, 8]. For this reason, character motion retargeting provides the necessary foundation for the present project, but must be extended to account for interaction-aware geometric relationships in more complex motion adaptation tasks.

2.2 Close Interaction Motion Adaptation

Close interaction motion adaptation extends conventional motion retargeting to settings in which two characters, or a character and the surrounding environment,

remain in close proximity and exhibit rich geometric relationships throughout the motion sequence [6, 7], as shown in Figure 2.

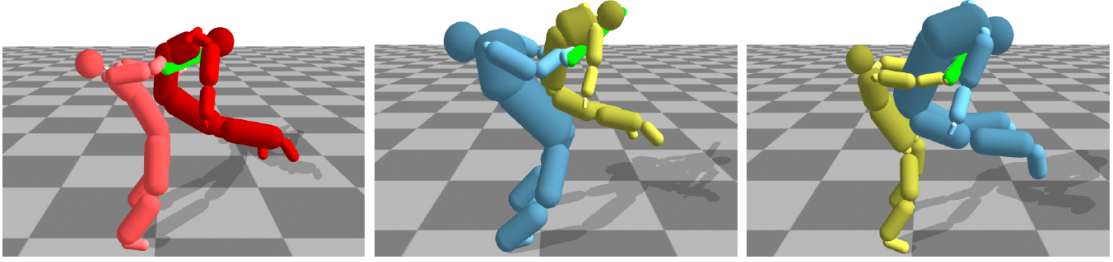


Figure 2: Back breaking attacking motion retargeted from left to middle and right. **Source:** Adapted from [6].

In such motions, the main objective is no longer limited to preserving the quality of each individual pose. Instead, the adapted motion must also retain the spatial and temporal structure of the interaction itself, since the semantics of behaviours such as dancing, wrestling, hugging, or fighting depend strongly on how the bodies move relative to one another over time [6, 8, 11].

This makes close interaction adaptation substantially more difficult than standard single-character retargeting. Prior work has shown that, while single-body retargeting can often be handled using geometric or dynamic constraints on one skeleton, two-character interaction requires preserving additional inter-character geometric and topological relationships, especially when the motion contains continuous contact, tangling, or coordinated close-body movement [6, 11].

As a result, optimization-based solutions for close interaction adaptation typically involve more complicated objective functions and a larger set of constraints, including positional, collision-related, and interaction-preserving terms. These methods can produce convincing adapted motions, but they are often computationally expensive and may require careful tuning when applied to a single motion sequence under new body configurations [6, 7].

At the same time, more recent data-driven work suggests that learning-based approaches may help address some of these limitations, but also highlights that two-person interaction introduces difficulties not present in single-body retargeting, particularly in preserving inter-character geometric structure under varying body sizes and limited data availability [11].

For this reason, close interaction motion adaptation has become an important and distinct problem within character animation. It builds on the foundations of conventional retargeting, but places much stronger emphasis on preserving interaction-aware spatial structure, which is central to the direction taken in this project [6, 8].

2.3 InteractionMesh and Spatial Relationship Preservation

A central idea behind InteractionMesh-based motion adaptation is that close interaction should be represented not only by the motion of individual joints, but also by the spatial relationships between nearby body parts and interacting objects or characters [6].

To achieve this, Ho et al. introduced the *Interaction Mesh*, a volumetric mesh constructed at each frame from the positions of character joints together with points sampled from interacting objects or the surrounding environment [6]. Rather than relying solely on joint angles and manually specified kinematic constraints, this representation aims to encode the local geometric context of the interaction directly in the motion data [6].

The key construction step is to treat these points as a single point cloud and apply Delaunay tetrahedralization, so that body parts or objects that are close to one another are naturally connected through mesh edges [6]. In this way, the resulting mesh captures those spatial relationships that are most relevant to the interaction, especially between nearby and non-occluded parts of the scene [6].

Based on this representation, motion adaptation is formulated as a deformation problem. The main objective is to preserve the local geometric details of the Interaction Mesh while the character bodies are edited or retargeted, which is achieved by minimizing Laplacian deformation energy over the mesh together with additional spacetime terms for temporal coherence [6]. In the original framework, this optimization is performed subject to constraints such as bone-length constraints, collision constraints, and positional constraints, so that the adapted motion preserves both the context of the interaction and the plausibility of the resulting body motion [6].

An important design choice in this framework is to represent posture using joint positions rather than joint angles. Ho et al. argue that this makes the constraint matrix sparser, because the joints can be treated as independent particles, whereas a joint-angle representation leads to dense dependencies along the kinematic chain [6]. For body segments whose orientation cannot be determined reliably from joint positions alone, the framework further introduces virtual vertices sampled on bounding volumes to retain additional orientation information [6].

This representation is particularly relevant to the present project because it provides a concrete bridge between geometric interaction modelling and constrained optimization. In later work, the same InteractionMesh formulation was shown to support multiresolution acceleration for close character interaction editing [7], and

was also adapted to motion synthesis in constrained robotic environments, where preserving implicit spatial relationships between a body and nearby obstacles is similarly important [5]. For this reason, InteractionMesh serves as the most direct technical foundation for the re-implementation carried out in this project.

2.4 Motion Representation and BVH-Based Skeleton Data

A suitable motion representation is essential for close interaction adaptation, since the adaptation process depends not only on the motion itself but also on how the geometric relationships between body parts are recovered from the input data. In this project, the motion data is represented in BVH format, which provides a hierarchical skeleton structure together with motion parameters defined over time.

For the purpose of interaction-aware motion adaptation, the most important information is the frame-wise spatial configuration of the skeleton. Although BVH stores motion through a hierarchy of joints, offsets, and local rotations, the later stages of the pipeline require explicit joint positions in Euclidean space, since spatial relationships, geometric constraints, and interaction structures are all defined in terms of positions rather than raw rotational channels.

This choice is also consistent with the InteractionMesh framework. Ho et al. represent body posture using joint positions instead of joint angles, arguing that this leads to a sparser constraint matrix because joints can be treated as independent particles, whereas joint-angle formulations introduce dense dependencies along the kinematic chain [6]. For close interaction adaptation, this is particularly useful because many of the relevant constraints, such as positional control, collision handling, and interaction-preserving geometric relationships, are more naturally expressed on points in space than on joint-angle coordinates [6].

Based on this representation, BVH data serves as the input to a preprocessing stage that reconstructs the skeleton hierarchy and computes frame-wise joint positions for each interacting character. These positions then form the basis for the construction of the InteractionMesh-style representation and the constrained adaptation formulation used later in the project.

At the same time, the use of skeleton-based motion data reflects a practical trade-off. Skeletal motion is compact and widely used in animation and motion analysis, but for close two-person interaction it does not directly encode all geometric or contact relationships by itself. More recent work on two-person interaction data therefore emphasises that, beyond the skeleton alone, the relations between joints across two bodies are crucial for describing the interaction semantics under varying body sizes

[11]. This further motivates the conversion from raw BVH motion data to a more interaction-aware geometric representation in the present project.

2.5 Learning-Based Solvers for Constrained Optimization

Recent work at the intersection of machine learning and optimization has explored whether neural networks can be used to approximate the input–output mapping of parametric optimization problems, thereby producing fast approximate solutions for families of related instances [4, 10]. This idea is attractive in settings where the same underlying optimization problem must be solved repeatedly under varying parameters, but where classical solvers may be too slow for interactive or near-real-time use [2, 4, 1].

From this perspective, a neural network can be viewed as a proxy solver that predicts candidate solutions directly from problem parameters [4, 9, 10]. However, a central difficulty immediately arises: while deep models are expressive and fast at inference time, their outputs do not naturally satisfy hard equality and inequality constraints, so naive regression-based approaches often produce infeasible solutions even when they approximate the objective well [2, 4, 9].

A common line of research therefore focuses on integrating constraint handling into the learning process itself. Some methods train neural networks with penalty-based losses that include constraint violation terms, often in supervised or unsupervised form, so that the model learns to trade off objective quality against feasibility [1, 12, 4]. Other approaches draw more explicitly on constrained optimization methods such as Lagrangian duality or augmented Lagrangian schemes, using neural networks together with learned multipliers or dual variables in order to reduce constraint violations more systematically [9, 4].

More broadly, recent surveys describe this area as part of end-to-end constrained optimization learning or optimization learning, where the goal is not merely to assist a solver, but to train differentiable proxy models that return fast approximate solutions while preserving as much feasibility and solution quality as possible [10, 4]. Within this broader literature, an important distinction is whether constraints are only encouraged through penalties, or whether the architecture includes explicit repair, completion, projection, or correction mechanisms that enforce feasibility more directly [4, 2].

For the present project, this line of work is relevant because the adaptation of close interaction motion is naturally formulated as a constrained optimization problem. Once the motion adaptation task is expressed in terms of geometric objectives and

hard constraints, it becomes reasonable to ask whether part of the solution process can be learned by a neural network rather than solved entirely by iterative optimization. This does not remove the need for constraints; rather, it shifts the problem toward designing a learning-based solver that remains compatible with the underlying constrained formulation.

2.6 DC3 and Its Relevance to This Project

Among recent learning-based approaches for constrained optimization, DC3 (Deep Constraint Completion and Correction) is particularly relevant to the present project because it addresses a central difficulty of neural proxy solvers: how to produce outputs that remain compatible with hard constraints [2].

The key idea of DC3 is that a neural network should not necessarily predict the full optimization solution directly. Instead, the network first predicts only a partial set of decision variables, after which the remaining variables are completed so as to satisfy the equality constraints [2]. This *completion* step is followed by a differentiable *correction* procedure that iteratively reduces inequality constraint violations while remaining on the manifold defined by the equality constraints [2]. In this way, DC3 differs from approaches that only encourage feasibility through penalty terms in the loss function, since it explicitly incorporates feasibility-oriented operations into the solution process itself [2, 4].

This distinction is important for the present project. The close interaction motion adaptation problem considered here is not merely an unconstrained regression task; rather, it is formulated with structured geometric objectives together with equality and inequality constraints, including constraints related to body configuration, interaction preservation, and collision-aware adaptation. A learning-based solver for this setting therefore needs to do more than predict visually plausible outputs: it should also remain consistent with the constrained optimization formulation underlying the adaptation process.

For this reason, DC3 provides a useful conceptual framework for integrating deep learning with the existing InteractionMesh-based formulation. Its emphasis on partial prediction, constraint completion, and constraint correction aligns well with the goal of learning candidate adapted poses while still respecting the geometric structure of the underlying optimization problem [2]. More generally, this is consistent with the broader idea of optimization learning, in which neural networks are combined with repair or completion mechanisms in order to produce fast but feasible proxy solutions for families of related optimization instances [4].

At the same time, the role of DC3 in this project should be stated carefully. The project does not attempt to reproduce the full theoretical scope of DC3 across generic optimization domains. Instead, DC3 is used here as the main methodological inspiration for the preliminary DNN-based solver, providing the constraint-aware design principle that learning should be coupled with explicit feasibility handling rather than treated as unconstrained motion prediction.

2.7 Dataset Source and Position of This Project

The motion data used in this project is closely related to the setting studied in *Two-Person Interaction Augmentation with Skeleton Priors*, which focuses on close and continuous two-body skeletal interactions under varying body sizes and proportions [11]. This is a suitable data source for the present project because it addresses the same general class of interaction motions, namely contact-rich or near-contact two-person motions in which preserving geometric relations between the two bodies is essential.

The use of this data source is also motivated by a broader limitation identified in the literature. As recent work has pointed out, high-quality skeletal datasets with close and continuous two-person interactions are relatively rare, and this lack of data makes both motion retargeting and learning-based interaction generation more difficult [11]. In particular, methods designed for single-body motion are generally insufficient for modelling the semantic and geometric structure of two-character interaction, especially when body size varies [11].

Within this context, the position of the present project is more limited and more specific. It does not aim to propose a new general interaction representation, nor does it attempt to develop a new generic framework for constrained optimization learning. Instead, the project lies at the intersection of these two directions. On one side, it re-implements the InteractionMesh-based formulation of spatial relationship preserving motion adaptation [6]; on the other side, it explores a preliminary integration of this formulation with a constraint-aware neural solver inspired by DC3 [2].

This means that the project should be understood primarily as an engineering reproduction and feasibility study. Its contribution is not a fully successful new method for close interaction motion adaptation, but rather the construction of a working pipeline that connects BVH-based motion data, InteractionMesh-style geometric modelling, constrained formulation, and preliminary DNN-based prediction within a single system.

For this reason, the project occupies a distinct position relative to the surrounding literature. Compared with the original InteractionMesh work, it focuses more on re-implementation and neural integration than on proposing a new geometric formulation [6]. Compared with recent learning-based approaches for constrained optimization, it focuses on a specific graphics and motion adaptation problem rather than on general optimization benchmarks [2]. Compared with recent two-person interaction generation work, it does not attempt to build a complete supervised generative model, but instead investigates whether an existing interaction-preserving optimization framework can be connected to a constrained neural solver in a practically meaningful way [11].

3 System Design and Methodology

3.1 Overall Pipeline Overview

The system developed in this project follows a pipeline that combines motion preprocessing, interaction-aware geometric modelling, constrained formulation, and preliminary neural prediction within a single workflow, shown in Figure 3. The input to

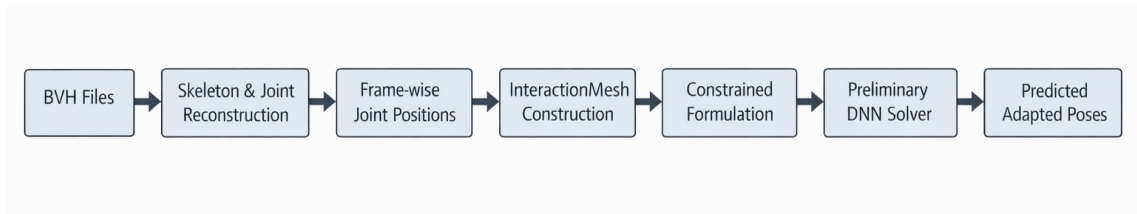


Figure 3: Overall pipeline of the proposed system.

the system consists of two-character motion sequences stored in BVH format. These files are first parsed to recover the skeleton hierarchy and the motion parameters of each character over time. From this information, frame-wise joint positions are reconstructed and used as the basic spatial representation for the later stages of the pipeline.

Based on the recovered joint positions, an InteractionMesh-style geometric representation is then constructed for the selected interaction motion. This step transforms the raw skeletal motion into a structure that explicitly captures spatial relationships between nearby body parts, which is the key geometric information required for close interaction motion adaptation [6].

Once this interaction-aware representation is available, the motion adaptation problem is formulated as a constrained optimization problem. The formulation combines

geometric preservation terms with several constraints that control the validity of the adapted motion, including positional constraints, bone-length related constraints, and collision-aware terms. This stage provides the optimization-based foundation of the system and defines the problem that the later learning-based solver attempts to approximate.

To enable learning-based prediction, the constrained formulation is further converted into a dataset construction process, in which multiple adaptation instances are generated under varying body size configurations. These instances are then used to train a preliminary neural solver that predicts candidate adapted poses from constraint-related inputs.

The neural solver is inspired by the completion-based idea of DC3 [2]. Rather than treating motion adaptation as an unconstrained prediction task, the system retains the structure of the underlying constrained problem and uses a constraint-aware prediction process to generate outputs that are more consistent with the optimization formulation.

Finally, the predicted outputs are converted back into a form suitable for qualitative inspection. In this project, the generated poses are exported and visualized in Blender in order to examine whether the adapted results preserve recognizable interaction structure under new character configurations.

Overall, the pipeline can be summarized as follows: BVH motion input, skeleton and joint reconstruction, InteractionMesh-style geometric construction, constrained motion adaptation formulation, dataset generation, preliminary DNN-based prediction, and qualitative visualization. The following sections describe each part of this workflow in detail.

3.2 BVH Parsing and Motion Representation

The input motion data used in this project consists of two-character interaction sequences stored in BVH format. Before any interaction-aware geometric modelling can be performed, the raw motion files must be converted into a representation that exposes the spatial configuration of the characters at each frame.

To achieve this, a custom BVH parser was implemented to read both the hierarchy section and the motion section of each file. The hierarchy parsing stage reconstructs the skeleton tree, including joint names, parent-child relationships, offsets, and end sites. In addition to the original skeleton nodes, virtual points are introduced for non-root joints in order to retain extra geometric information for later interaction-aware modelling. These virtual points are generated from the local joint geometry

and stored in the same tree structure as the original skeleton representation.

After the hierarchy has been reconstructed, the motion section is parsed frame by frame. For each frame, the local motion channels are converted into rotation matrices, and forward kinematics is applied recursively along the skeleton hierarchy to recover the global position of every joint. The resulting output is therefore not a sequence of raw BVH channels, but a dictionary of frame-wise 3D joint positions for each character, which is more suitable for the later stages of the pipeline.

This choice of representation is consistent with the InteractionMesh framework discussed in Chapter 2. Rather than operating directly on joint angles, the present system uses explicit joint positions as the main motion representation, since the subsequent geometric construction and constrained formulation are defined in Euclidean space. In particular, positional constraints, bone-length related terms, and interaction-aware geometric structures are all more naturally expressed on point coordinates than on local rotational channels [6].

The output of this stage is therefore a frame-indexed collection of skeletal point sets, together with the associated parent–child relationships extracted from the BVH hierarchy. This representation forms the basis for the InteractionMesh construction described in the next section. Algorithm 1 summarises the main parsing procedure used in the system.

Algorithm 1 BVH Parsing and Joint Position Reconstruction

Require: BVH file f , scaling factor s

Ensure: Frame-wise joint positions $\mathcal{F}$ and skeleton hierarchy $\mathcal{H}$

- 1: Read the BVH hierarchy and motion sections from f
 - 2: Reconstruct the skeleton hierarchy $\mathcal{H}$ from the hierarchy section
 - 3: Record joint names, offsets, end sites, and parent–child relations
 - 4: Add virtual points for non-root joints when required
 - 5: **for** each frame t in the motion section **do**
 - 6: Read root translation and local joint rotations
 - 7: Convert local Euler angles to rotation matrices
 - 8: Apply recursive forward kinematics along the hierarchy
 - 9: Store the global 3D position of each joint and virtual point in $\mathcal{F}[t]$
 - 10: **end for**
 - 11: **return** $\mathcal{H}, \mathcal{F}$
-

3.3 InteractionMesh Construction

Once the frame-wise joint positions have been reconstructed from the BVH motion data, the next step is to convert these skeletal point sets into an interaction-aware geometric representation. In this project, this is achieved by constructing an

InteractionMesh-style structure for each frame of the selected two-character motion sequence.

The purpose of this construction is to make the spatial relationships between nearby body parts explicit. A raw skeleton representation only provides a set of joints connected by the kinematic hierarchy, but it does not directly encode which parts of the two characters are close to each other in space at a given frame. For close interaction motion adaptation, this information is essential, since the semantics of the interaction depend not only on the posture of each character individually, but also on the relative arrangement of their body parts over time [6].

Following the original InteractionMesh framework, the reconstructed joint positions are treated as a point cloud, and a volumetric connectivity structure is derived from them using Delaunay tetrahedralization [6]. This construction produces tetrahedral connections between spatially nearby points, allowing the representation to capture local geometric relationships across the two interacting characters. In practice, the resulting tetrahedral structure is not used as a surface mesh for rendering, but as an interaction-aware geometric scaffold for defining deformation and constraint terms in the later optimization stage.

In addition to the original skeleton joints, the construction also includes auxiliary points associated with body segments. As described in the BVH parsing stage, virtual points are introduced for non-root joints in order to retain additional geometric information beyond the basic joint chain. These points are incorporated into the same frame-wise representation, so that the InteractionMesh captures not only the connectivity of the articulated skeleton, but also a richer local description of the spatial layout around each segment.

The output of this stage consists of several geometric structures used throughout the rest of the system. First, each frame is represented as a dictionary of point positions, including joints, end sites, and virtual points. Second, parent-child relationships in the skeleton are converted into explicit bone definitions and indexed bone pairs, which are later used for bone-length constraints and collision-related modelling. Third, the Delaunay tetrahedralization is reduced to a set of tetrahedral edges, which define the neighbourhood relationships required for the Laplacian-style deformation terms in the constrained formulation.

This construction is important because it provides the geometric bridge between motion representation and constrained adaptation. Instead of formulating the adaptation problem directly on isolated skeletal joints, the system operates on an interaction-aware structure that preserves neighbourhood information across both characters. As a result, later deformation and constraint terms can be defined on a representa-

tion that is much closer to the spatial logic of the original interaction than a purely kinematic skeleton alone.

In summary, the InteractionMesh construction stage transforms frame-wise joint positions into a set of point-based geometric structures that explicitly capture local proximity and articulated connectivity. These structures form the basis for the constrained motion adaptation formulation described in the next section. Algorithm 2 summarises the main construction procedure. Figure 4 shows the visualization of joint points, virtual points and highlighted tetrahedral edges in orange by Blender.

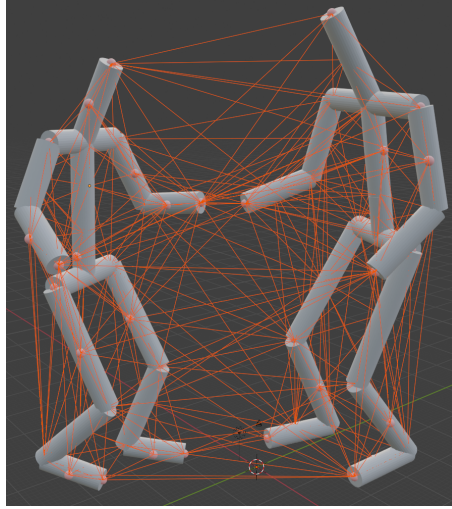


Figure 4: Visualization of Interaction Mesh. Joints and virtual points are visualized as red spheres and tetrahedral edges are orange lines.

Algorithm 2 InteractionMesh Construction from Frame-wise Joint Positions

Require: Frame-wise joint positions $\mathcal{F}$, parent–child pairs $\mathcal{P}$

Ensure: Interaction-aware geometric structures for all frames

- 1: **for** each frame t **do**
 - 2: Collect all joint, end-site, and virtual-point positions into a point set V_t
 - 3: Apply Delaunay tetrahedralization to V_t
 - 4: Extract tetrahedral edges and store them as neighbourhood relations
 - 5: **end for**
 - 6: Convert parent–child relations $\mathcal{P}$ into explicit bone definitions
 - 7: Build indexed bone pairs for later bone-length and collision modelling
 - 8: Store the frame-wise point sets, bone structures, and tetrahedral edges
 - 9: **return** all frame-wise geometric structures
-

3.4 Constrained Motion Adaptation Formulation

After the InteractionMesh-style geometric structures have been constructed, the close interaction adaptation task is formulated as a constrained optimization prob-

lem defined on the frame-wise point coordinates. For a given frame i , let n denote the number of points in the interaction-aware representation, and let

$$V_i \in \mathbb{R}^{3n}$$

be the stacked vector of all point coordinates in that frame. The goal is to compute an updated configuration

$$V'_i \in \mathbb{R}^{3n}$$

that preserves the geometric structure of the original interaction while satisfying the adaptation constraints.

3.4.1 Deformation Energy

The main objective term is a deformation energy derived from the InteractionMesh representation. Its role is to preserve the local geometric relationships of the original point set during adaptation. Following the Laplacian-style formulation, a weighted graph Laplacian is first constructed from the tetrahedral edge structure of the current frame. Let

$$L_i \in \mathbb{R}^{n \times n}$$

denote the random-walk Laplacian of frame i , and let

$$M_i = L_i \otimes I_3 \in \mathbb{R}^{3n \times 3n}$$

be its expansion to 3D coordinates, where I_3 is the 3×3 identity matrix.

The deformation energy is then defined as

$$E_{\text{def}}(V'_i) = \frac{1}{2} \|M_i(V'_i - V_i)\|_2^2.$$

This term penalises changes in the Laplacian coordinates of the interaction point set, and therefore encourages the updated configuration to preserve the local spatial structure of the original frame.

Expanding the quadratic form gives

$$E_{\text{def}}(V'_i) = \frac{1}{2} V_i'^\top Q_i V'_i + p_i^\top V'_i + \text{const},$$

where

$$Q_i = M_i^\top M_i, \quad p_i = -M_i^\top M_i V_i.$$

In the implementation, this quadratic term provides the main geometry-preserving

component of the adaptation objective.

3.4.2 Soft Constraints

In addition to the deformation term, the formulation includes soft constraints that guide the adapted configuration toward desired target behaviour without enforcing exact equality.

Positional constraint. A subset of points is selected as positional anchors, and their target positions are specified explicitly. In the current implementation, these anchor points are chosen from foot-related joints and end sites so that the adapted result remains more stable and visually plausible. Let

$$K_i \in \mathbb{R}^{3k \times 3n}$$

be the selection matrix that extracts the k constrained points from V'_i , and let

$$P_i \in \mathbb{R}^{3k}$$

be the stacked target positions. The positional energy is defined as

$$E_{\text{pos}}(V'_i) = \frac{1}{2} \|K_i V'_i - P_i\|_2^2.$$

This term softly encourages the selected points to move toward the specified target locations.

Penetration-related term. To reduce interpenetration between body segments, each bone is approximated as a line segment with an associated radius. For pairs of non-adjacent bones, the closest points between the two segments are computed. When the distance between the segments is smaller than the sum of the radii, a penetration relation is detected. These relations are assembled into a linear form

$$J_i V'_i \approx f_i,$$

where J_i is a penetration matrix and f_i is the corresponding target vector. This leads to a soft penetration energy

$$E_{\text{pen}}(V'_i) = \frac{1}{2} \|J_i V'_i - f_i\|_2^2.$$

In the current implementation, this term provides a formulation for collision-aware adaptation, although collision handling is not yet fully integrated into the final learned solver.

Combining the above components, the frame-wise objective can be written in the general form

$$E_i(V'_i) = E_{\text{def}}(V'_i) + \lambda_{\text{pos}} E_{\text{pos}}(V'_i) + \lambda_{\text{pen}} E_{\text{pen}}(V'_i),$$

where λ_{pos} and λ_{pen} are weighting coefficients controlling the relative influence of the soft terms.

3.4.3 Hard Constraints

The formulation also includes hard constraints, whose purpose is to preserve structural validity during adaptation.

Bone-length constraint. Let $\mathcal{B}$ denote the set of bones in the skeleton, represented as parent-child point pairs. For each bone, the target length is determined from the scaled character configuration. In the implementation, the bone-length constraints are linearised around the current configuration, resulting in a matrix form

$$A_i V'_i = b_i,$$

where A_i is constructed from the local bone directions and b_i contains the target bone lengths. This equality constraint ensures that the updated configuration remains consistent with the desired skeletal proportions.

In the current system, the final adaptation problem for each instance is therefore expressed as a quadratic objective together with equality constraints of the form

$$\min_{V'_i} \frac{1}{2} V_i'^{\top} Q_i V'_i + p_i^{\top} V'_i \quad \text{subject to} \quad A_i V'_i = b_i,$$

with optional additional soft terms such as positional and penetration energies included in the objective.

3.4.4 Equality Completion for the Neural Solver

To integrate this constrained formulation with a neural network solver, the equality constraints are handled using a completion-based strategy inspired by DC3 [2]. Rather than predicting the entire vector V'_i directly, the network predicts only a

subset of variables. The remaining variables are then recovered so that the equality constraints are satisfied.

Suppose the equality system is

$$A_i y = b_i,$$

where y denotes the full optimization variable. If the columns of A_i can be permuted into

$$A_i = [A_{\text{partial}} \ A_{\text{other}}],$$

with A_{other} invertible, then the variable vector can be partitioned as

$$y = \begin{bmatrix} y_{\text{partial}} \\ y_{\text{other}} \end{bmatrix}.$$

Given a predicted y_{partial} , the remaining variables are completed by

$$y_{\text{other}} = A_{\text{other}}^{-1}(b_i - A_{\text{partial}} y_{\text{partial}}).$$

This allows the final output to satisfy the equality constraints by construction.

In the implementation, an invertible block is identified using QR decomposition with column pivoting. When such a block exists, the neural solver predicts only the unconstrained part of the variable vector, and the rest is recovered analytically through equality completion. This design allows the learning module to remain compatible with the constrained optimization structure, rather than treating motion adaptation as unconstrained regression. Inequality-based correction, which would be required for stronger collision-aware feasibility, is left for future work.

3.5 Dataset Construction

In order to train a neural solver for motion adaptation, the constrained formulation described in the previous section is converted into a dataset of parametric optimization instances. Rather than treating the problem as a standard supervised learning task with manually annotated target poses, the dataset is constructed directly from the optimization formulation itself. Each sample corresponds to one adaptation instance of the form

$$\min_y \frac{1}{2} y^\top Q y + p^\top y \quad \text{subject to} \quad A y = x, \quad G y \leq h,$$

where the matrices Q , p , A , G , and h define the structure of the adaptation problem, x represents the instance-specific hard constraint constants, and y denotes the

adapted point configuration to be predicted.

3.5.1 Reference Frame and Base Formulation

The dataset is constructed from a selected frame of a two-character interaction sequence. In the current implementation, a single interaction motion is chosen as the reference example, and one frame from that motion is used to build the base optimization formulation. From this frame, the current point configuration V_{current} is extracted, together with the corresponding InteractionMesh edge structure, skeleton bones, and indexed bone pairs.

Using this reference configuration, the deformation term is assembled first. The tetrahedral edge structure of the selected frame is used to construct the Laplacian matrix, which then defines the quadratic objective

$$\frac{1}{2}y^\top Qy + p^\top y,$$

with

$$Q = M^\top M, \quad p = -M^\top b,$$

where M is the Laplacian weight matrix and $b = MV_{\text{current}}$ is the vector of original Laplacian coordinates. This objective is shared across all generated instances, since it captures the geometry-preserving preference of the reference interaction.

3.5.2 Construction of Equality Constraints

The equality constraints are divided into two parts: positional constraints and bone-length constraints. These are combined into a single matrix

$$A = \begin{bmatrix} H_1 \\ H_2 \end{bmatrix},$$

where H_1 is the positional selection matrix and H_2 is the linearised bone-length constraint matrix.

For the positional part, a set of foot-related joints and associated end or virtual points is selected as fixed anchors. This design choice is motivated by the observation that fixing the feet yields more stable and visually plausible retargeted results than constraining arbitrary points. The target values of these anchor points are collected into a vector of fixed positions.

For the bone-length part, the skeleton bones are first extracted from the parent–

child structure, and bones whose endpoints are both already fixed by the positional constraints are removed from the bone-length constraint set. The remaining bones are then used to construct the matrix H_2 , which encodes the local bone directions and linearises the target bone-length conditions.

3.5.3 Instance Generation by Bone-Length Scaling

The key source of variation in the dataset comes from changing the target bone lengths. Instead of constructing only one optimization problem, multiple problem instances are generated by applying random scaling factors to the reference bone lengths. Let l denote the vector of bone lengths in the reference frame. For each generated instance, a scaling vector s is sampled and the target bone-length constants are modified accordingly.

In this way, each sample represents a different body-size configuration while preserving the same underlying interaction structure. The fixed positional targets remain unchanged, whereas the desired skeletal proportions vary from sample to sample. The resulting hard constraint constants are stacked into the instance vector

$$x = \begin{bmatrix} x_{\text{pos}} \\ x_{\text{bone}} \end{bmatrix},$$

where x_{pos} contains the fixed point locations and x_{bone} contains the scaled bone-length targets.

This construction yields a family of related optimization instances that share the same geometric objective and constraint structure, but differ in their target body proportions. In the current implementation, a large number of such instances are generated by randomly sampling bone scaling factors within a prescribed range, producing the input matrix

$$X \in \mathbb{R}^{N \times d_x},$$

where each row corresponds to one instance-specific constraint vector x .

3.5.4 Inequality Constraints

If penetration relations are detected in the reference configuration, they are assembled into an inequality system

$$Gy \leq h,$$

where G is the penetration matrix and h is the corresponding target vector. If no such penetration terms are detected, the inequality system is left empty. In

the present implementation, this inequality component is included in the dataset representation, although it is not yet fully exploited by the final learning pipeline due to the absence of a complete correction-based solver.

3.5.5 Input and Output of the Neural Solver

This dataset construction also determines the learning interface. The input to the neural network is the instance vector x , i.e. the concatenation of the hard constraint constants for a particular adaptation configuration. The output is the adapted variable vector y , which represents the updated frame-wise point coordinates.

This means that the network is trained to learn the mapping

$$x \mapsto y,$$

where the input specifies the desired constraint targets and the output corresponds to a feasible or near-feasible adapted pose under the fixed quadratic formulation. For each generated instance, the input vector x contains the hard constraint constants, including fixed-point targets and scaled bone-length targets, while the output vector $y \in \mathbb{R}^{3m}$ represents the adapted 3D coordinates of all points in the frame. Because the dataset is generated directly from the optimization model rather than from manually labelled target poses, the learning process is better interpreted as optimization-driven or self-supervised training than as conventional supervised pose regression.

Overall, the dataset construction stage converts a single reference interaction frame into a family of constrained adaptation problems with varying target body proportions. This provides the neural solver with a structured set of training instances while keeping the geometric objective and interaction-preserving formulation fixed.

3.6 Preliminary DNN-Based Solver

After the constrained adaptation problem has been converted into a dataset of parametric optimization instances, a neural solver is introduced to approximate the mapping from instance-specific constraint constants to adapted poses. The purpose of this module is not to replace the geometric formulation, but to learn a fast approximation to the solution process while remaining compatible with the underlying constrained structure.

3.6.1 Network Architecture

The current solver uses a relatively simple multilayer perceptron (MLP). The input dimension is determined by the number of instance-specific constraint constants in x , and the output dimension is determined by the dimension of the optimization variable y . In the implementation, the network consists of two hidden layers, each followed by batch normalization, ReLU activation, and dropout. A final linear layer produces the predicted variables.

This design was chosen deliberately. Since the main purpose of the present project is to test the feasibility of integrating constrained optimization with neural prediction, a compact MLP is sufficient as a first solver architecture. The focus is therefore placed on the interaction between network prediction and constraint handling, rather than on the use of a highly sophisticated motion generation architecture.

3.6.2 Training Objective

The neural solver is trained using the optimization objective together with soft penalties on constraint violation. Let $\hat{y}$ denote the predicted output after the forward pass and, where used, equality completion. The training loss is defined as

$$\mathcal{L}(\hat{y}) = f(\hat{y}) + \lambda_{\text{ineq}} \|\text{ineq_dist}(x, \hat{y})\|_2 + \lambda_{\text{eq}} \|\text{eq_resid}(x, \hat{y})\|_2,$$

where $f(\hat{y})$ is the quadratic objective value, `ineq_dist` measures inequality violation, and `eq_resid` measures equality residual. In the implementation, the relative weight between the equality and inequality penalties is controlled by a hyperparameter that determines how the total soft constraint weight is split between the two terms.

This loss makes the training process optimization-driven rather than purely supervised. The network is not trained only to imitate a target pose; instead, it is encouraged to produce outputs with low objective value and low constraint violation under the original motion adaptation formulation. The main process is better shown in the Algorithm 3.

3.6.3 Role of Correction

Although the code framework includes gradient-based correction procedures for both training and testing, the current implementation mainly relies on equality completion and does not yet make full use of a stable correction-based pipeline. In particular, inequality-aware correction was intended to improve feasibility with respect to collision-related terms, but this component was not fully integrated into the final

Algorithm 3 Forward Pass of the Preliminary DNN-Based Solver

Require: Instance-specific constraint vector x

Ensure: Predicted adapted pose $\hat{y}$

- 1: Feed x into the multilayer perceptron
- 2: Obtain the predicted partial variables $\hat{y}_{\text{partial}}$
- 3: **if** equality completion is enabled **then**
- 4: Recover the remaining variables using

$$\hat{y}_{\text{other}} = A_{\text{other}}^{-1}(x - A_{\text{partial}}\hat{y}_{\text{partial}})$$

- 5: Concatenate $\hat{y}_{\text{partial}}$ and $\hat{y}_{\text{other}}$ to form $\hat{y}$
 - 6: **else**
 - 7: Set $\hat{y}$ to the full network output
 - 8: **end if**
 - 9: **return** $\hat{y}$
-

system.

For this reason, the present solver should be understood as a completion-based preliminary neural solver rather than a full reproduction of DC3. Its main contribution is to demonstrate that the motion adaptation problem can be connected to a constraint-aware learning pipeline, while leaving more complete correction-based feasibility handling for future work.

3.7 Prediction and Visualization Workflow

After training, the neural solver is used to generate adapted poses for new body-size configurations by following a separate prediction workflow. This stage reuses the same constrained formulation and data construction logic as in training, but applies them to a newly specified adaptation setting rather than to a batch of randomly generated dataset instances.

3.7.1 Construction of a New Test Instance

To perform prediction, a new input instance is first constructed from the same reference interaction frame used in dataset generation. The fixed positional targets are kept unchanged, while the target bone lengths are modified according to the desired scale factors of the two characters. In the current implementation, separate scaling factors can be assigned to the two bodies, allowing a new adaptation scenario to be created from the same interaction motion.

Using these target values, a new hard-constraint vector x_{new} is assembled in the same format as the training data. This means that the prediction stage remains

fully consistent with the formulation used during training: the neural solver receives the same type of input as before, but corresponding to a user-specified body-size configuration rather than to a randomly sampled dataset instance.

3.7.2 Forward Prediction

Once the new input vector has been constructed, it is passed through the trained neural solver in evaluation mode. The output is a predicted adapted pose represented as a stacked variable vector, which is then reshaped into frame-wise 3D point coordinates. Since the same completion-based forward pass is used as in training, the predicted output remains tied to the equality-constrained formulation of the adaptation problem.

In practice, this stage produces a deformed point configuration for the selected interaction frame under the new target body proportions. The predicted coordinates serve as the main output of the solver and form the basis for both numerical inspection and later visualization.

3.7.3 Numerical Checking

Before visual inspection, the predicted output is checked numerically against the constrained formulation. In particular, equality residuals are computed in order to measure how closely the predicted pose satisfies the equality constraints. These statistics include mean and maximum absolute residuals, which provide a simple diagnostic of whether the completion-based solver remains close to the intended constrained structure.

Additional numerical checks are also carried out on specific geometric quantities. Since the main source of variation in the adaptation task is the target bone lengths, the predicted bone lengths are compared with the reference ones to examine whether the expected scaling behaviour is reflected in the output. Similarly, fixed-point locations associated with the feet are compared before and after deformation to verify whether the positional anchoring is approximately preserved.

These numerical tests do not fully determine motion quality, but they are useful for identifying obvious failure modes before visual analysis. They also provide a direct link between the learned output and the original formulation in Chapter 3, since they test whether the predicted pose behaves consistently with the intended hard constraints.

3.7.4 Export to Blender and Qualitative Inspection

Although the prediction output is numerical, the final quality of close interaction adaptation is easier to assess visually than from constraint values alone. For this reason, the predicted 3D coordinates are exported as NumPy arrays together with the corresponding bone indices and tetrahedral edge structure. These files are then loaded into Blender for qualitative inspection.

A dedicated Blender script reconstructs the predicted structure by first transforming the coordinates into Blender’s axis convention, then building a point-edge mesh from the predicted vertices and skeleton connectivity. The bone structure is visualized using cylinders placed between indexed point pairs, and the tetrahedral neighbourhood edges can optionally be displayed as an additional wireframe overlay.

This visualization serves two purposes. First, it makes it possible to inspect whether the two-character interaction remains recognizable after adaptation. Second, it provides a more intuitive way to identify geometric distortions, implausible proportions, or interaction failures that may not be obvious from numerical residuals alone. In the present project, Blender is therefore used not as part of the solver itself, but as a practical inspection tool for evaluating the predicted poses.

3.7.5 Role in the Overall Pipeline

Taken together, the prediction and visualization workflow forms the final stage of the system pipeline. The trained neural solver produces an adapted pose for a new constraint configuration, numerical testing checks its consistency with the constrained formulation, and Blender visualization provides a qualitative assessment of whether the interaction structure is preserved. This workflow bridges the gap between the internal optimization variables of the solver and the interpretable outputs discussed in the experimental evaluation chapter.

3.8 Implementation Scope and Current Limitations

The system described in this chapter should be understood as a working prototype rather than a complete solution to close interaction motion adaptation. Its main scope is to build a full engineering pipeline that connects BVH-based motion representation, InteractionMesh-style geometric modelling, constrained adaptation formulation, dataset construction, and preliminary neural prediction within a single framework.

Within this scope, the implementation successfully reconstructs frame-wise inter-

action geometry from BVH data, formulates the adaptation task as a constrained optimization problem, and generates a family of parametric adaptation instances with varying target body proportions. It also demonstrates that a neural network can be integrated with the constrained formulation through equality completion, so that prediction is no longer treated as unconstrained pose regression.

At the same time, the current system has several important limitations. First, the learning-based solver is preliminary in nature and does not yet provide a fully reliable solution across all adaptation settings. Although the equality constraints are handled through completion, inequality-aware correction is not fully integrated into the final pipeline. As a result, collision-related feasibility is represented in the formulation but not yet enforced by a complete correction-based mechanism during prediction.

Second, the implementation focuses on adaptation at the level of a selected reference frame and its derived optimization instances, rather than on full-sequence motion generation. This means that the current system should be interpreted as solving a frame-wise pose adaptation problem within an interaction-aware setting, rather than synthesizing a temporally complete retargeted motion sequence.

Third, several design choices in the current implementation are intentionally pragmatic. For example, foot-related points are used as positional anchors because they lead to more stable and visually plausible outputs, and the range of body scaling factors is chosen to generate meaningful but controlled variations around the reference interaction. These choices make the system easier to analyze and test, but they also mean that the current implementation is not intended to cover all possible interaction configurations or constraint strategies.

Finally, although the predicted outputs can retain recognizable interaction structure, they do not always satisfy all intended geometric properties perfectly. In particular, exact preservation of target body proportions and bone lengths remains imperfect in some cases. For this reason, the present implementation is best viewed as a feasibility study of constraint-aware neural motion adaptation built on top of an InteractionMesh-based formulation, rather than as a finished retargeting system.

These limitations are important not only as caveats, but also as indicators of where future improvements are needed. In particular, more complete inequality handling, stronger geometric consistency, and extension from frame-wise adaptation to temporally coherent motion generation remain the main directions for further development.

4 Results and Evaluation

4.1 Evaluation Setup

The evaluation in this project focuses on assessing whether the proposed pipeline can preserve interaction structure under character scaling, while approximately satisfying the imposed constraints. The evaluation is conducted on a representative motion sequence and consists of both qualitative inspection and quantitative analysis.

The motion data used in this evaluation is an “around-the-back” interaction sequence extracted from the dataset introduced in [11]. This sequence involves close physical interaction between two characters, making it suitable for evaluating whether spatial relationships are preserved after motion adaptation.

Due to the scope of this project, the evaluation is performed on a single representative frame (frame 100) rather than the full motion sequence. This frame-wise evaluation is sufficient for the present study, as the primary focus is on geometric constraint satisfaction and interaction structure preservation, rather than temporal coherence across frames.

To generate adapted motions, character-specific scaling is applied to bone lengths. During dataset construction, scaling factors are sampled within the range $[0.75, 1.25]$ to create diverse training examples. For evaluation, two representative test cases are selected:

- Case 1: $sf_A = 1.0$, $sf_B = 0.75$ (one character scaled down)
- Case 2: $sf_A = 0.8$, $sf_B = 1.2$ (asymmetric scaling)

To ensure physical plausibility, a subset of joints corresponding to the feet of both characters are fixed during adaptation. This prevents unrealistic global translation or floating effects and provides stable positional anchors for the optimization process.

The evaluation is carried out using two complementary perspectives:

- **Qualitative evaluation:** visual inspection of the adapted poses in Blender to assess whether interaction structure is preserved.
- **Quantitative evaluation:** analysis of bone-length ratios to measure how well the target scaling constraints are satisfied.

It is worth noting that equality constraints are enforced through the completion step in the DC3-inspired framework, and are therefore satisfied by construction at

inference time. During training, however, equality residuals are still included as soft penalties to stabilise learning. As a result, equality constraints are not used as an evaluation metric in this project, and the analysis instead focuses on geometric consistency and constraint approximation in the final outputs.

4.2 Qualitative Results

This section presents qualitative results of the proposed pipeline under different character scaling configurations. The goal is to evaluate whether the interaction structure between two characters is preserved after motion adaptation.

Figure 5 shows three representative cases: the original motion, and two adapted motions under different scaling settings.

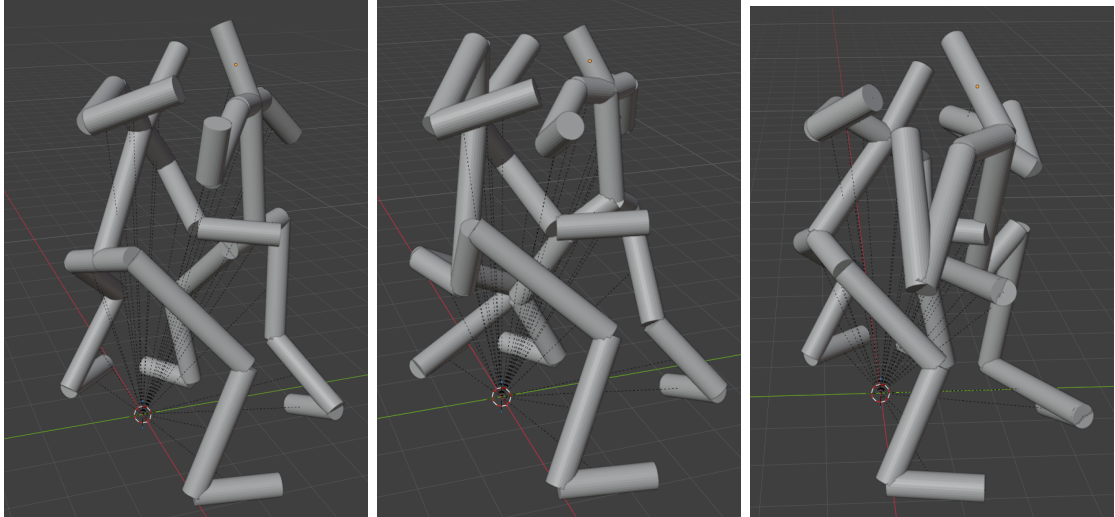


Figure 5: Qualitative results of close interaction motion adaptation. Left: original motion. Middle: $sf_A = 1.0$, $sf_B = 0.75$. Right: $sf_A = 0.8$, $sf_B = 1.2$.

In the original motion (left), the two characters exhibit a close interaction pattern, where their limbs are spatially coordinated and positioned relative to each other. This serves as the reference configuration for evaluating the adapted results.

In the first adaptation case ($sf_A = 1.0$, $sf_B = 0.75$, middle), character B is scaled down while character A remains unchanged. The adapted result shows that the overall interaction structure is largely preserved. The relative positioning between the two characters remains consistent with the original motion, and the proximity between corresponding body parts is maintained. This suggests that the proposed method is capable of adapting motion across moderate scale differences while retaining the semantic relationship between interacting characters.

In the second adaptation case ($sf_A = 0.8$, $sf_B = 1.2$, right), the scale difference

between the two characters becomes more pronounced. In this scenario, the global interaction pattern is still recognizable, indicating that the method captures high-level spatial relationships. However, local distortions become more evident, particularly in limb alignment and joint orientation. Some body parts appear less naturally positioned, suggesting that the deformation does not fully satisfy all geometric constraints.

Overall, these qualitative results demonstrate that the proposed pipeline is able to preserve the general interaction structure under character scaling. At the same time, they also reveal limitations in maintaining precise geometric consistency, especially under larger or asymmetric deformations.

4.3 Bone-Length Scaling Analysis

To quantitatively evaluate how well the adapted poses satisfy the intended scaling constraints, the ratio between each deformed bone length and its original length is computed:

$$r_i = \frac{l'_i}{l_i}, \quad (1)$$

where l_i and l'_i denote the original and adapted lengths of bone i , respectively. Ideally, r_i should match the target scaling factor applied to the corresponding character.

For each evaluation case, the bone-length ratios are divided into two groups according to the two characters. The mean ratio, standard deviation, and mean absolute error (MAE) with respect to the target scaling factor are then computed. The results are shown in Table 1.

Table 1: Bone-length scaling analysis under two evaluation settings.

Case	Character	Target scale	Mean ratio	MAE to target
(1.0, 0.75)	A	1.0	1.088	0.088
(1.0, 0.75)	B	0.75	0.848	0.098
(0.8, 1.2)	A	0.8	0.829	0.029
(0.8, 1.2)	B	1.2	1.572	0.372

The results show that the proposed pipeline is able to approximately follow the desired scaling trend in some cases. In particular, for the (1.0, 0.75) setting, the mean bone-length ratios remain reasonably close to the target values for both characters, indicating that the solver can partially preserve the intended body proportions under moderate deformation.

However, the more asymmetric setting (0.8, 1.2) reveals a significant limitation of the current system. While Character A remains close to the target scale, Character B exhibits a much larger deviation, with a mean ratio of 1.572 compared with the

target value of 1.2. This indicates that the present solver does not reliably enforce bone-length constraints when the required deformation becomes stronger.

Overall, these results suggest that the current implementation can capture the general direction of target scaling, but does not yet achieve stable and accurate bone-length preservation across all adaptation settings.

4.4 Discussion of Failure Cases and Limitations

The qualitative and quantitative results presented in the previous sections reveal that, although the proposed pipeline is capable of preserving the overall interaction structure, it does not fully satisfy all geometric constraints. This section discusses the main failure cases and analyzes the underlying reasons.

A primary limitation lies in the incomplete enforcement of bone-length constraints. As shown in Table 1, the adapted bone lengths deviate from the target scaling factors, especially in the (0.8, 1.2) case. In particular, Character B exhibits large errors and high variance, indicating that the solver is unable to maintain consistent bone proportions under stronger asymmetric deformation. This issue is also visible in the qualitative results, where some limbs appear stretched or misaligned.

This behaviour can be attributed to the formulation of the problem as a soft-constrained optimization combined with a learning-based approximation. In the current implementation, bone-length constraints are incorporated into the constrained formulation, but the neural solver does not explicitly solve the optimization problem. Instead, it learns an approximate mapping from constraint parameters to adapted poses. While equality constraints are handled through completion, this does not guarantee that all geometric constraints are simultaneously satisfied.

Another important limitation is the absence of explicit correction for inequality constraints, such as collision avoidance. Although a penetration constraint is formulated, it is not actively enforced during training or inference through a correction procedure. This limits the ability of the system to resolve conflicts between geometric constraints, especially in close interaction scenarios where body parts are in proximity.

Furthermore, the current approach relies on a purely data-driven approximation of an optimization process. While the neural network is able to learn a mapping from constraint parameters to adapted poses, it does not explicitly solve the underlying optimization problem. This makes the method sensitive to distribution shifts, such as larger scaling differences that are not well represented in the training data. The instability observed in the (0.8, 1.2) case reflects this limitation.

The current solver should therefore be understood as a learned approximation of the constrained optimization process, rather than a replacement for exact optimization. This distinction is important for interpreting both the strengths and limitations of the system.

Finally, the evaluation itself is limited in scope. The results are based on a single motion sequence and a single frame, which does not capture temporal consistency or generalization across different interaction types. Therefore, while the current results provide useful insights into the behaviour of the system, they should be interpreted as a preliminary validation rather than a comprehensive evaluation.

Despite these limitations, the failure cases are informative. They highlight the key challenges in combining learning-based methods with constrained motion adaptation, particularly the difficulty of enforcing multiple interacting constraints simultaneously. These observations provide clear directions for future improvements, such as incorporating full correction steps in DC3, improving constraint balancing, and extending the evaluation to more diverse motion data.

5 Conclusion and Evaluation

This project set out to explore close interaction motion adaptation using a combination of classical constraint-based formulation and a learning-based solver. The primary goal was not to fully solve the problem, but to construct a working pipeline and investigate the feasibility of integrating deep neural networks into a constrained optimization framework.

The project successfully implemented a complete pipeline, starting from BVH motion data and progressing through skeleton reconstruction, InteractionMesh construction, constrained problem formulation, dataset generation, and a preliminary DNN-based solver. The system is able to generate adapted poses under different character scaling configurations, and qualitative results demonstrate that the overall interaction structure can be preserved to a certain extent.

However, the evaluation results also show that the system does not fully satisfy all geometric constraints. In particular, bone-length consistency is not reliably maintained, especially under stronger and asymmetric scaling. While the network is capable of approximating the target deformation, it struggles to enforce multiple constraints simultaneously. This indicates that the current approach is not yet sufficient for producing physically or geometrically consistent motion adaptation.

From a design perspective, the project highlights several important challenges. First,

formulating motion adaptation as a constrained optimization problem introduces inherent trade-offs between different types of constraints. Second, learning-based solvers, while efficient, do not naturally guarantee feasibility. Third, the absence of a full correction mechanism limits the ability to resolve constraint violations after prediction.

Despite these limitations, the project provides several valuable contributions. It demonstrates that the InteractionMesh-based formulation can be successfully re-implemented as an end-to-end pipeline, making the original method more accessible and inspectable. It also shows that neural networks can partially approximate the behaviour of constrained solvers, suggesting a promising direction for accelerating motion adaptation. Perhaps most importantly, the project identifies the key technical difficulties that must be addressed in order to make such approaches reliable.

There are several directions for future work. A natural extension is to incorporate the full correction step of the DC3 framework, which could improve constraint satisfaction by refining the predicted outputs. Another direction is to improve the balance between different constraints, for example by introducing adaptive weighting schemes or alternative formulations. Extending the evaluation to multiple motion sequences and temporal consistency would also provide a more comprehensive assessment of the method. Finally, exploring more advanced network architectures or hybrid optimization-learning approaches may lead to more robust solutions.

In conclusion, while the current system does not fully solve close interaction motion adaptation, it provides a useful engineering foundation and a preliminary validation of learning-based approaches under constraints. The insights gained from the failure cases are particularly valuable, as they clarify the challenges of combining optimization and deep learning in this domain.

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